

Adroit Technologies

Protocol Driver

Name	GSM SMS Serial
DLL	GSM_SMS



COPYRIGHT

Copyright © 2022 Advanced Worx 112 (Pty) Ltd. All rights reserved.

Information in this document is subject to change without notice. The software described in this document is furnished under a license agreement or nondisclosure agreement. The software may be used or copied only in accordance with the terms of those agreements. No part of this publication may be reproduced, stored in a retrieval system, or transmitted in any form or any means electronic or mechanical, including photocopying and recording for any purpose other than the purchaser's personal use without the written permission of Advanced Worx 112 (Pty) Ltd.

Advanced Worx 112 (Pty) Ltd
20 Waterford Office Park
189 Witkoppen Road (c/o Waterford Drive)
Fourways
South Africa

<https://adroittech.co.za/>

Contents

1. Overview	4
2. Wiring	5
3. Device Configuration	6
3.1. Driver Installation and Setup.....	6
3.1.1. Devices.....	6
3.1.2. Device Configuration Dialog	6
3.1.3. GSM_SMS Driver Details Settings	6
3.1.4. Using Cost Control	7
3.1.5. Using a SIM PIN	9
3.1.6. Unicode Languages	9
4. Supported Addresses	10
5. Protocol Definition and Considerations	11
5.1. Using the Notify Agent with GSM_SMS Driver.....	11
5.1.1. Phase 1: Creating the Agents.....	11
5.1.2. Phase 2: Configuring Alarming	11
5.1.3. Phase 3: Alarming.....	12
5.1.4. Phase 4: Scanning	12
5.2. Configuring a Notify Agent.....	12
5.3. Configuring a Notify Agent.....	15
5.4. Message Limitations	15
5.5. Tag Greying/Tags marked bad	15
5.6. Limitations	16
5.7. Message Protocols.....	17
5.7.1. Start-up	17
5.7.2. General Poling for SMS's.....	17
5.7.3. Sending a SMS	18
5.7.4. Receiving a SMS from an Unknown Source.....	18
6. General Notes and Troubleshooting	19
6.1. Contact Details.....	19
6.2. Adroit Technologies Knowledge Base.....	19
6.3. Troubleshooting	19
7. Appendix A – Abbreviations.....	20
8. Revisions.....	20
8.1. Document Revision.....	20
8.2. DLL Revision	20
9. Appendix B - Unicode (Multi-byte) Languages (Send only)	22

1. Overview

The GSM_SMS protocol driver for Adroit using GSM modems. Messages configured in String or Notify Agents may be sent via SMS. Alarms that occur in Adroit can thus be routed to a person via SMS. The driver only handles the transmission of the intended message, which can originate as an alarm, a message entered by an operator or from programming done in Adroit. This driver is intended for easy use with any cellular SMS protocol. No Windows drivers for the modem are needed, this driver provides its own interface to the modem via the serial port assigned to it.

All tests and designs were carried out using a 'WAVECOM WMOD2' model GSM Modem.



PLEASE NOTE

This driver does not currently support secondary ports.

2. Wiring

Standard depending on configuration. Can be utilised with RS-485, RS-232, Modem, or Ethernet.

3. Device Configuration

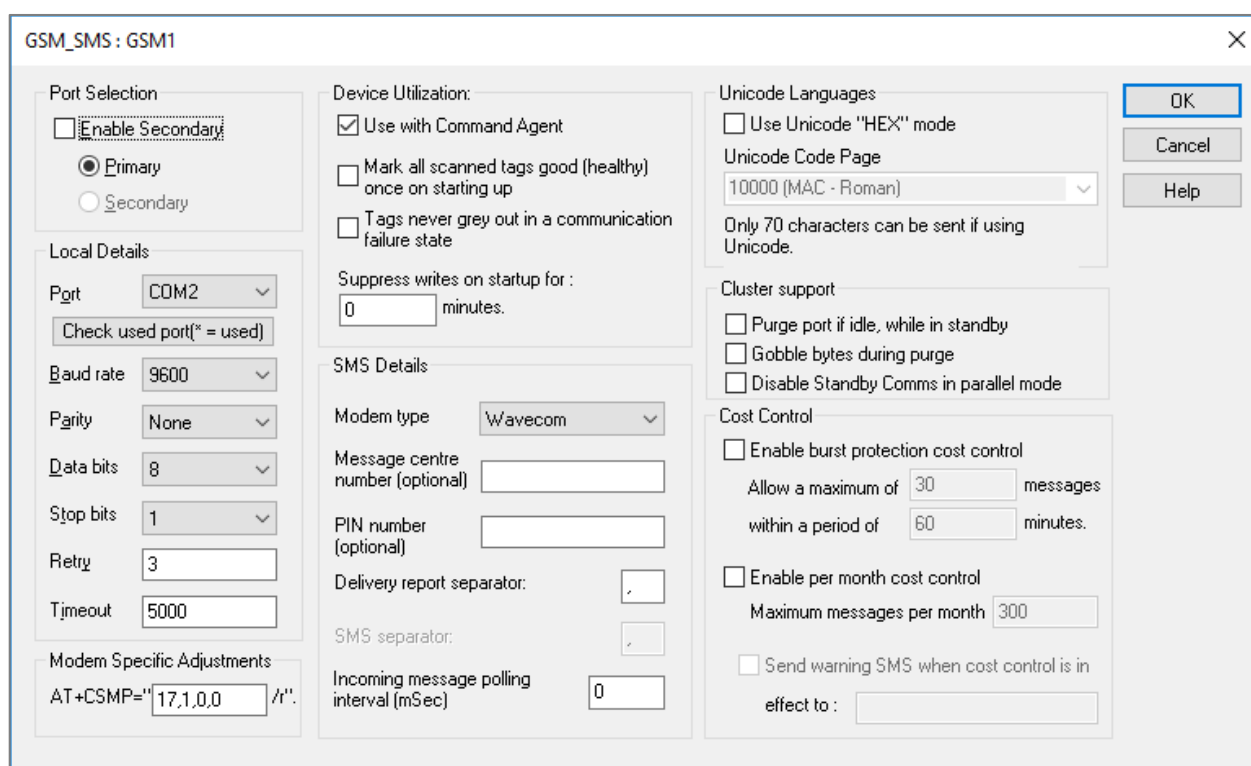
3.1. Driver Installation and Setup

Installation of the GSM_SMS protocol driver is done in the standard way in Adroit Setup. After installation of the driver, the user may then create and configure devices for the GSM_SMS driver.

3.1.1. Devices

At a minimum, you must create at least one GSM_SMS device for each modem. A new device must be created per service provider. Currently, the port settings for all devices on a port must be identical. In other words, it is not permitted for example to use Vodacom and MTN together.

3.1.2. Device Configuration Dialog



3.1.3. GSM_SMS Driver Details Settings

Port	The communication port to which the modem/device is attached.
Baud rate	The speed of data transmission. Any Baud Rates may be specified from 110 to 256000.
Retries	Number of attempts to communicate.
Timeout	Number of milliseconds before a transaction will be deemed as timed out. This setting is a safety in case the remote device keeps giving the same error response.
Use with Command Agent	If checked, then the device will only work for Command Agent Tags.
Mark all scanned tags good (healthy) on starting up	By default all tags are scan-bad or grey on start-up, selecting this option will enable all the tags on start up. However, if the tags then fail to communicate after this, they will be displayed as scan-bad (grey).

Tags never grey out in a communication failure state	This option prevents tags from being marked bad. WARNING: If this option is used, then your Device Agent will NEVER show that it is unhealthy, even if the GSM modem stops working.
Suppress writes on startup for x minutes.	Default:0 – normal operation, if > 0 on start-up will cause the driver to suppress and consume any SMS alarm sending until this period expires, only new alarms after this period will be sent as normal.
Modem type	Type of GSM modem used. Currently ONLY the "Wavecom" option is supported, which is the default selection. Theoretically other modems should work as the driver attempts to use the standard GSM Hayes command set.
Message centre number	This is the message centre number, which will overwrite the number in your SIM. Example: – MTN +27831000002 – Vodacom +27829119 – Cell-C +27841000000
PIN Number (optional)	This is optional because it is not necessary to enable and use a SIM PIN. See below for the advantages and disadvantages of using a SIM PIN and how to configure this parameter in both cases.
Delivery Report Separator	This is the .CSV delimiter, defaulted to your locale setting. This may be changed as required. (default = comma)
SMS Separator	This setting is reserved for future use and is network provider dependent. However, the comma separator has worked on all the cell networks encountered so far.
Incoming message polling interval (mSec)	This setting affects the update rate of the signal strength parameter. It is recommended to use a minimum interval of 1000 milliseconds, since smaller values have no effect.
Modem Specific Adjustments	Some modems does not all support the standard AT+CSMP=17,1,0,0 command and thus allows the user to change it accordingly.
Enable per month cost control	This allows a user to cap the number of messages that can be sent in a month. WARNING: Use this at your own risk, any messages sent after the cap has been reached will be discarded and you will never receive them.
Maximum messages per month	The number of messages that can be sent in a month. Note the month starts from the 1 st of the specific month.
Enable burst protection cost control	This allows a user to cap the number of messages that can be sent in a specific time period. For example, if your plant goes into an alarmed state, use this to prevent multiple SMS's being sent which all refer to the same alarm. WARNING: Use this at your own risk, any messages exceeding the maximum allowed within the period of time will be discarded.
Allow a maximum of X messages	This is the maximum number of messages that can be sent in the period specified below.
Within a period of X minutes	This is the period in minutes in which messages will be capped if they exceed the maximum specifies above.
Purge port if idle while in standby	This is a hard purge (an API call) it has been found to sometimes cause locking (depending on the device used). If this occurs, see the point below.
Gobble bytes during purge	This is a soft purge; it is a standard read to clear the buffer. This has been found to be more tolerant than a hard purge on certain devices.
Disable standby comms when in parallel mode	Turn this on if your cluster system Is in parallel mode and you have reads or writes enabled in standby. This will disable all comms to the modem from the standby server.

3.1.4. Using Cost Control

When using either of the cost control features, be aware that SMS's will be discarded if the limits are reached. With that in mind, be sure that suitable limits are set, i.e., avoid setting the maximum allowable messages too low and avoid setting the time period too long.

3.1.5. Using a SIM PIN

The advantage of using a SIM PIN is that if someone on site steals the SIM card, they will NOT have full access to making calls. However, this does slow down the initialization process, and may require manual PUK unblocking if the wrong PIN is entered more than 3 times.

If using a SIM PIN:

Enter the required PIN number in the **PIN Number** edit box, which is stored in the registry, which is encrypted. Then when connecting to the modem, the first request carried out is - SIM PIN status and if this is required, we use the one provided.

If you want to not specify a SIM PIN:

In this case, it is necessary to unlock your SIM card, as follows:

1. Install the SIM card into a mobile phone.
2. Use the menu functions (which different per phone) to access the security settings and turn off SIM checking.
3. Remove the SIM card from the phone and install it into the GSM modem.

3.1.6. Unicode Languages

Use Unicode "HEX" mode	This will enable the sending of Multi-byte strings using Unicode code pages. Will adjust the +CSMP string slightly x,x,x,8 or x,x,x,0 if disabled. (Default is unchecked)
Unicode Code page ComboBox	This ComboBox will be populated with installed and available code pages. Select the page required.

4. Supported Addresses

You may only use String type slots.

'+' (Read Write)

For sending and receiving messages, use the international style phone number. For example, +27821234567

'\$DR' (Read only)

For delivery Reports you must use '\$DR' followed by the number as above (no spaces etc.). This will give you notification of your SMS when it is delivered to that phone number.

'\$UNKNOWN_MSG' (Read only)

Any SMS that is received from an unknown destination (not scanned) will be updated to the String slot scanned with this address.

'\$UNKNOWN_NUMBER' (Read only)

The sender's number will be sent to this Sting Slot.

Example:

If you configure 4 String Agents viz. SHIFT_MANAGER, SHIFT_MANAGER_DR, UNKNOWN_MSG, UNKNOWN_NUMBER, and scan them to the addresses...

Tag Name	Address
SHIFT_MANAGER.value	+27830000000
SHIFT_MANAGER_DR.value	\$DR+27930000000
UNKNOWN_MSG.value	\$UNKNOWN_MSG
UNKNOWN_NUMBER.value	\$UNKNOWN_NUMBER

Messages can be sent and received to and from the Shift Manager, with delivery notification, to ensure that the Operator knows that the Shift Manager has received his SMS.

If another operator is running late for his shift, he can send an SMS, and it will be received by the UNKNOWN_MSG with his number sent to UNKNOWN_NUMBER (if there are no tags set up for this operator).

5. Protocol Definition and Considerations

5.1. Using the Notify Agent with GSM_SMS Driver

A very useful feature of the GSM_SMS driver is its ability to route Adroit information (alarms, values, statuses, messages etc.) to cellphones. Adroit makes use of the Notify Agent to do this. The GSM_SMS driver will receive text messages from Adroit via the Notify Agent and send. The format of the text (string) message that the Notify Agent sends the GSM_SMS driver is configurable. It can be a simple analogue value, a string, or as is most useful, actual Adroit Alarms. This section explains how to set up the routing of an Adroit Alarm to a cellphone, using the Notify Agent. **You will create on Notify Agent for each user that needs to be sent any messages.**

First, we shall assume that we wish to alarm an Analog Tag. This feature is not limited to Analogs; any Adroit tag type may be alarmed in its place.

5.1.1. Phase 1: Creating the Agents

Step 1

Create a new Analog Agent. Edit the Analog Tag and configure High and Low alarms as required.

Step 2

Create a new AlarmList Agent. Call it something meaningful.

Step 3

Create a new Alarm Agent. Call it something meaningful.

Step 4

Create a new Notify Agent. Edit the Notify Agent and ensure that the "User Code" is left empty. Check the box entitled "Transmit an Alarm when it is raised". We shall assume the default alarm information is to be transmitted in the text message in this example. If you wish to have control over the format of this message, however, you may select which alarm fields you wish to transmit by pressing the "Format" button and editing the Alarm Record Structure.

5.1.2. Phase 2: Configuring Alarming

Step 5

Edit the Alarm Agent. You will automatically be in the "Routes" tab. It is imperative that you select a route other than zero. Route 0 is the default route used for general alarms. Under Output Destinations, go to the edit box entitled "Alarm Lists". Select the AlarmList Agent you created in Step 2.

Step 6

Go to the edit box entitled "Output". Select the Notify Agent you created in Step 4.

Step 7

Click on the "Types" tab. In the "Agent Type" drop-down box, select the Analog tag. Now in the "Available Alarm Types" edit box, remove the "DefaultAlarmAgent" from the box.

Step 8

Now we add some more types. When you click on the "Add" button, a new dialog will appear. In

the "Alarm Type" edit box, write the name of the alarm you wish to configure (e.g. ANA_1_HI). Now click the checkbox entitled "High" (for our example with the high alarm). Then, from the list-box entitled "Available:" move the RawHi alarm to the "Current" list-box.

Step 9

Repeat Step 8 for all the alarms you wish to have routed to a cellphone.

5.1.3. Phase 3: Alarming

Step 10

Now select the Analog Agent from the configurator and click on the "Alarm" button on the configurator. The Alarm Types configured in steps 8 and 9 should appear in the "Available" edit box. Remember to select the AlarmList Agent configured in Step 1. Now move the Alarm Types from the "Available" edit box to the "Current" edit box. Select the route number you configure these alarms via. The "Output" edit box should now list the Notify Agent you configured in Step 4. This means that all the Analog alarms of the types appearing in "Current" will be routed via the route number selected, to the Notify Agent appearing in the Output edit box.

5.1.4. Phase 4: Scanning

Step 11

Ensure that the tag is output enabled.

Scan the Notify Agent's ***rawValue*** slot, to the phone number, ensuring that you include the international dialling code.



WARNING

Mobile number must be entered in international format. For example, if the number to be dialled is 082 123 4567, then the format is as follows:

+2782123456

I.e., international format must be used.

5.2. Configuring a Notify Agent

To use a COMMAND Agent that uses the GSM_SMS driver with ESCALATION/ACKs:

Edit Command agent : C1

General
Phone
Pseudonyms and Macros
Subscriptions

Phone Settings:
☒ Use 'GSM_SMS' Device

GSM1

Recipient phone number:

+27825606235

☐ Allow anonymous callers
Cmd phone numbers:

+

-

SMS Delivery Confirmation:
☐ Require delivery report

Timeout (in minutes) 2

☒ Require recipient response

Timeout (in minutes) 1

Resend attempts before escalation:

2

Escalation Tag: (agent name only)

C2

'GSM_SMS' Device Diagnostics:

Signal Strength: 58

29 % or less is considered to be unreliable!

Raw Value

*2020/08/26 16:04:06 Valve ON

☐ Resend message

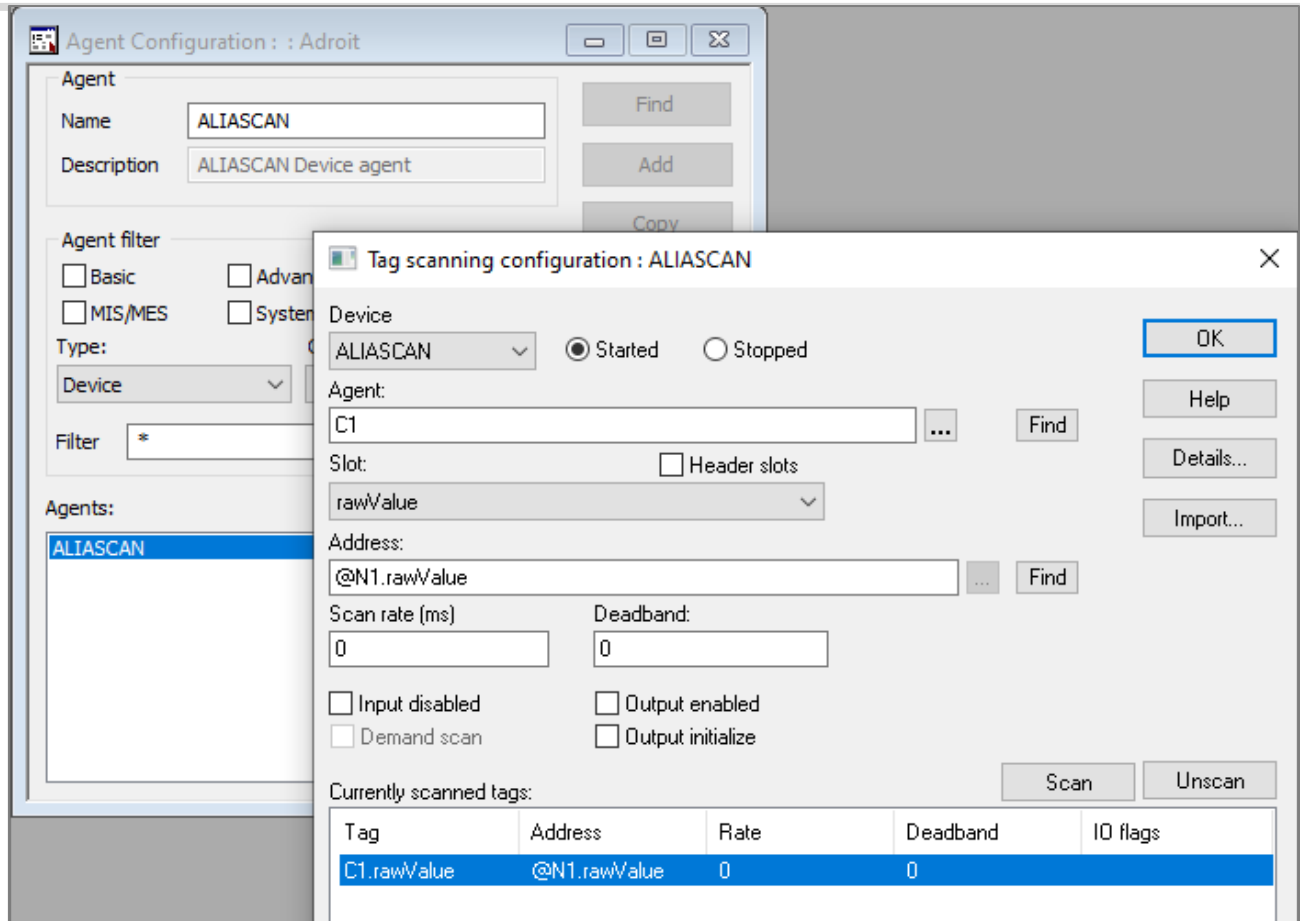
OK
Cancel
Update
Refresh
Header
Help

Setup a Command Agent (which represents a user), set the Agent to use GSM_SMS device mode, add the name of a valid GSM device.

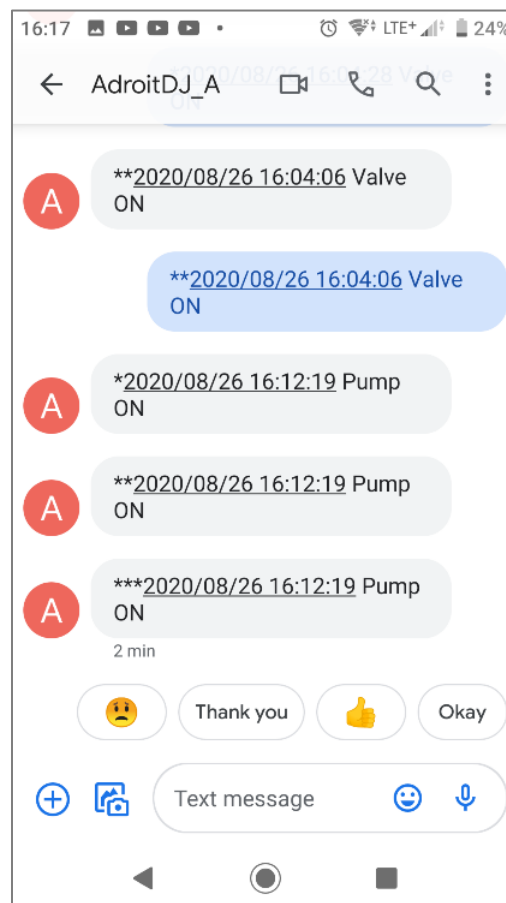
Add the users phone number, allow for additional functions like delivery reports, and user responses that can act as an Acknowledge mechanism for escalation cancellation purposes.

In this case C2 will be sent the message if the user configured in C1 does not respond by sending a copy of the received text back to sender in a reply within 1 minute (timeout) and 2 additional 1-minute period, then it resends.

Link the output of the text source (Agent) via @Scanning:

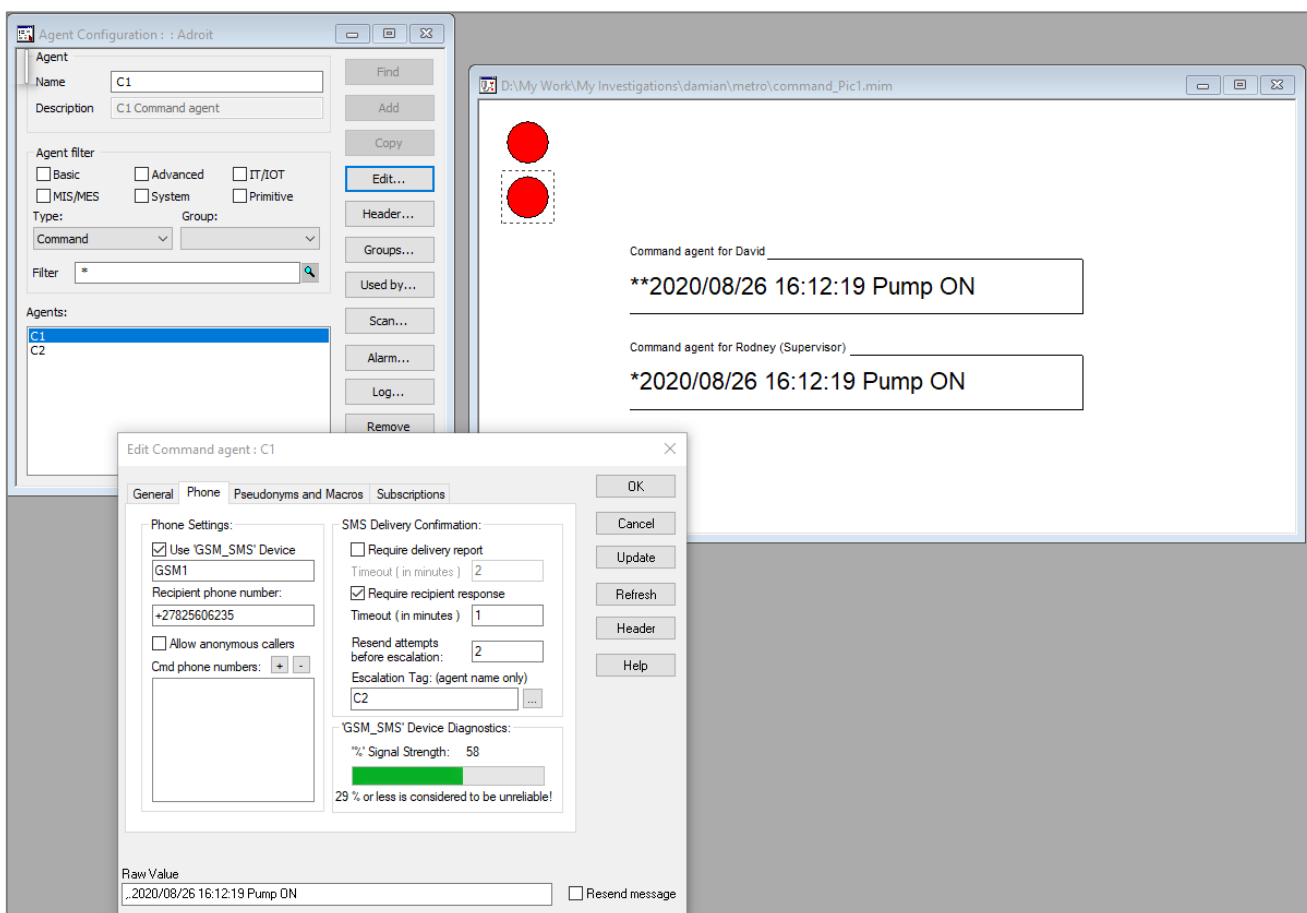


For example: Notifier N1 generates the output for Command C1, we "@Scan C1's raw value to/@ N1's raw value.



The blue reply of the exact text can be seen above...

The next 3 sends prefixed with "*" "*" "*" donate the first and 2 subsequent resend attempts, after the C1 user does not reply, the message is send to the C2 (escalation) user.



5.3. Configuring a Notify Agent

The Adroit Notify Agent must be configured in Adroit as above, (see Adroit manual - section: Notify Agent) and must be **output enabled**. In the case of Notify Agents, the slot called "User Code" must be left blank when using the GSM_SMS driver.

The Notify Agent can also support User entered messages. Simply type the message into the value slot from an Operator Action, "Set value" and the message will be sent when clicking on the OK button.

Note: to re-send an old message, you must change the message content by, for example, adding a space at the end of it.

5.4. Message Limitations

Limitations may apply to the content of messages; the driver will NOT convert illegal characters in a message. Generally, only use keyboard characters. The message length is limited to 79 characters, including the device code and a comma added for internal purposes when using the Notify Agent. If the Command Agent is being used, then SMS's will be a maximum of 160 characters.

5.5. Tag Greying/Tags marked bad

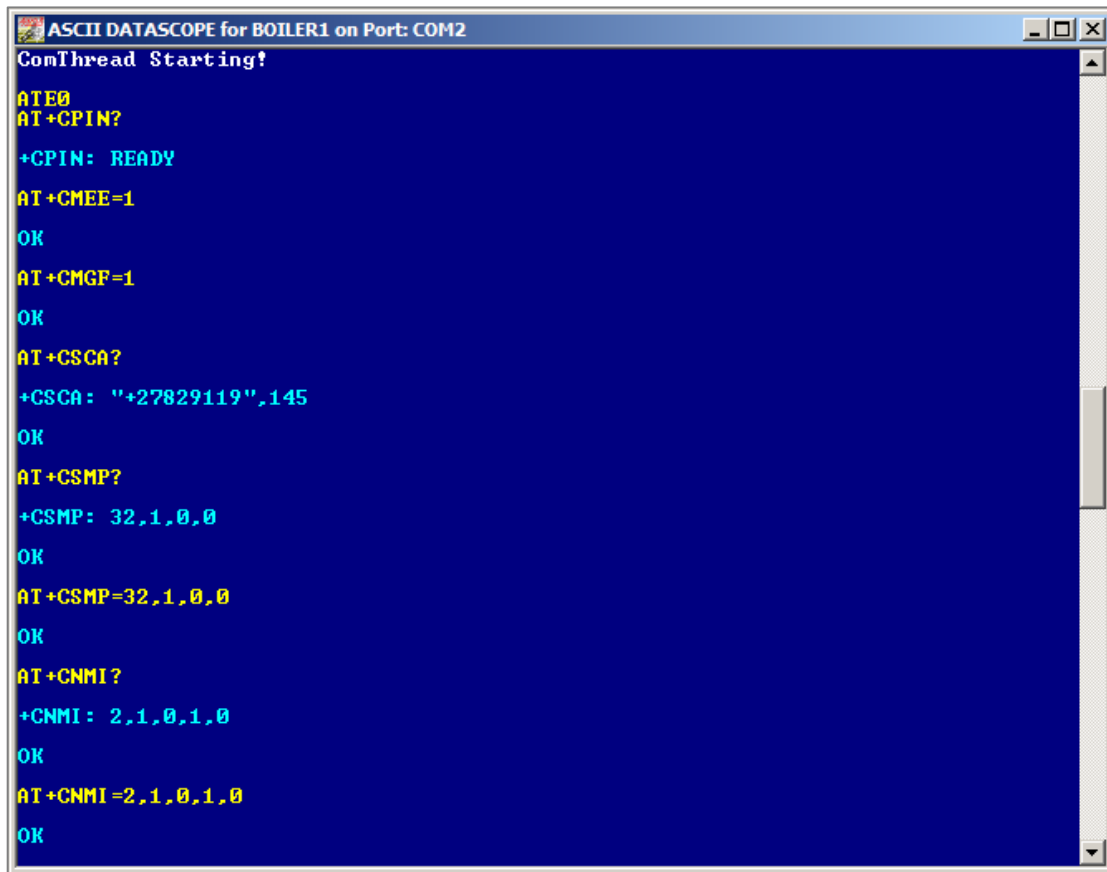
Scanned strings will not be greyed in the normal fashion used to show communications failures. Specify a non-zero scan-time to ensure that tags are always shown as enabled, even after a failure the tag will stay healthy. See the table of script commands for other ways to stop greying from occurring.

5.6. Limitations

- The driver works in Text mode only, and not PDU.
- Limited to 78 characters using Notify Agent.
- Limited to 160 characters using the Command Agent.

5.7. Message Protocols

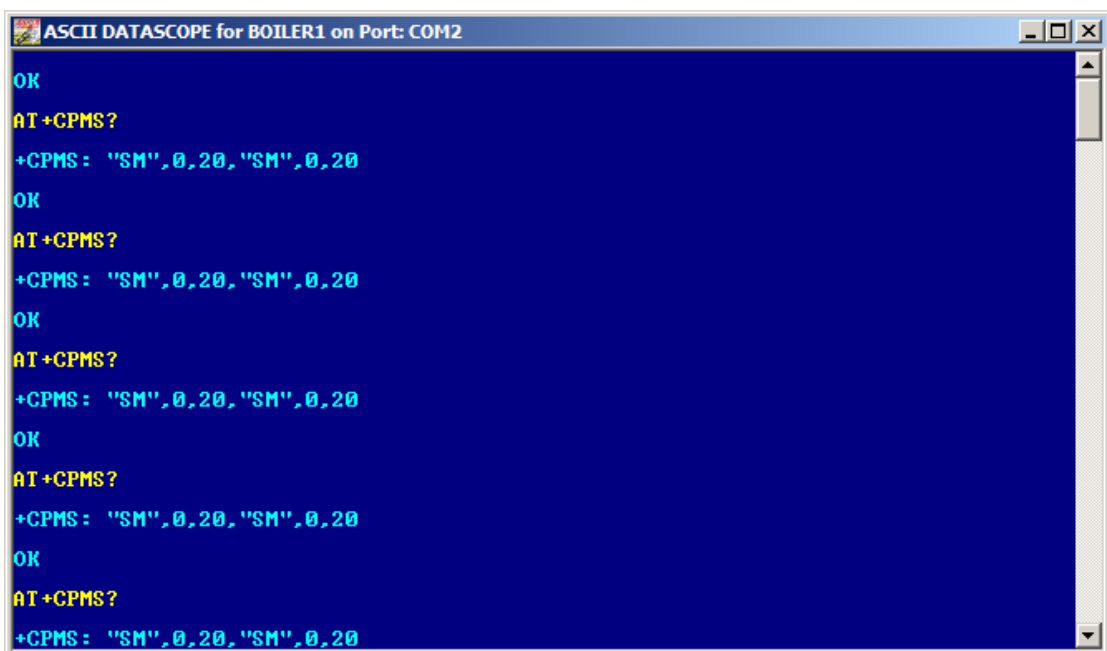
5.7.1. Start-up



```

ASCII DATASCOPE for BOILER1 on Port: COM2
ComThread Starting!
ATE0
AT+CPIN?
+CPIN: READY
AT+CMEE=1
OK
AT+CMGF=1
OK
AT+CSCA?
+CSCA: "+27829119",145
OK
AT+CSMP?
+CSMP: 32,1,0,0
OK
AT+CSMP=32,1,0,0
OK
AT+CNMI?
+CNMI: 2,1,0,1,0
OK
AT+CNMI=2,1,0,1,0
OK
  
```

5.7.2. General Poling for SMS's



```

ASCII DATASCOPE for BOILER1 on Port: COM2
OK
AT+CPMS?
+CPMS: "SM",0,20,"SM",0,20
OK
AT+CPMS?
+CPMS: "SM",0,20,"SM",0,20
OK
AT+CPMS?
+CPMS: "SM",0,20,"SM",0,20
OK
AT+CPMS?
+CPMS: "SM",0,20,"SM",0,20
OK
AT+CPMS?
+CPMS: "SM",0,20,"SM",0,20
  
```

5.7.3. Sending a SMS

```

ASCII DATASCOPE for BOILER1 on Port: COM2
AT+CPMS?
+CPMS: "SM",0,20,"SM",0,20
OK
[WRITING] [TO: 27830000000F] [MSG: Hello World]
AT+CMGS="+27830000000"
>
Hello World+
+CMGS: 187
OK

[MSG REFERENCE No: 187]
AT+CPMS?
+CPMS: "SM",0,20,"SM",0,20
OK
AT+CPMS?
+CPMS: "SM",0,20,"SM",0,20
OK
AT+CPMS?
+CPMS: "SM",0,20,"SM",0,20
OK
AT+CPMS?
+CPMS: "SM",0,20,"SM",0,20

```

5.7.4. Receiving a SMS from an Unknown Source

```

ASCII DATASCOPE for BOILER1 on Port: COM2
AT+CPMS?
+CMII: "SM",1
+CPMS: "SM",1,20,"SM",1,20
OK
AT+CMGL="ALL"
+CMGL: 1,"REC UNREAD","+27830000000","04/08/16,15:41:54+08"
new message sent
OK
AT+CMGD=1
OK

[Decoded SMS Deliver recieved from : +27830000000]
new message sent
[Updating Adroit] Device : BOILER1 : CellNumber : $UNKNOWN_MSG
[Updating Adroit] Device : BOILER1 : CellNumber : $UNKNOWN_NUMBER
AT+CPMS?
+CPMS: "SM",0,20,"SM",0,20
OK
AT+CPMS?

```

6. General Notes and Troubleshooting

6.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroit.co.za

6.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

[Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback \(adroittechnologiesautomation.com\)](http://adroittechnologiesautomation.com)

6.3. Troubleshooting

What do I do if the device's response for a step is random or undefined?

1. Use an empty RX step, the driver will read and display the response without doing anything else with it.
2. The PFF script is re-loaded every time the datascope displays [Attempt n/n Sending xxxx] or [Acquire I/O data], certain script variables require a restart, reload or change.

7. Appendix A – Abbreviations

DT	Dial Tone
DTR	Date Terminal Ready (a hardware signal)
GSM	Global System for Mobile telecommunications
PIN	Personal Identification Number
PUK	Personal Unblocking Key
SMS	Short Message Service

8. Revisions

8.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
1.2	19-Jan-2023	K. Robinson	Grammar edits, formatting changes and content additions	D. Jordaan
3.0				
4.0				
5.0				
6.0				
7.0				

8.2. DLL Revision

Rev.	Date	Changes
1	16/08/2004	Initial release
27	27/08/2020	Added Escalation resend count and maintenance work, removed “.” Prefixes that were used for the deprecated PAGER driver. NB – please use Rev 20 of the Command agent available from 10.0.5 onwards.
31	31/08/2022	+CPMS? was working but the driver kept repeating back to ATE0, fixed.

9. Appendix B - Unicode (Multi-byte) Languages (Send only)

When using Unicode languages there are some points to consider.

Due to the Hex mode used on the modem only half the amount of characters can be sent. i.e 70.

The correct code page must be selected, if incorrect the message will likely be garbled. It is usually best to have the computers local setting match the code page.

English can be sent simultaneously due to the fact that the English alphabet is always accessible, but again due the use of hex mode the 70 character limit still applies.

If a code page is visible in the Unicode Code page combo box in setup , that means it is a valid and may be used, the driver will acquire the installed and compatible codepages automatically.

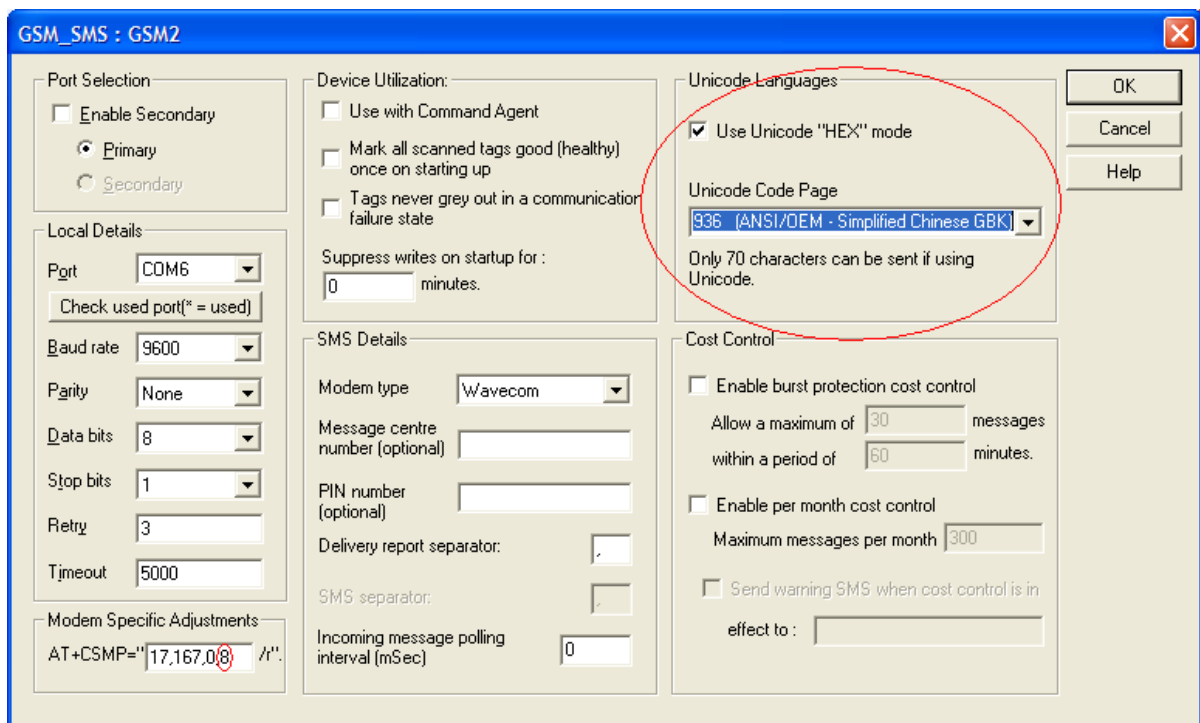
Not all code pages are clearly marked, in those cases an online search may prove use full. For example : Polish is a Latin based language and its special characters are available on ISO-8859-2 Latin-2 code page.

When selecting Hex mode, the 4th value of the +CSMP string will automatically be changed to 8 from 0 and vice versa.

The GSM_SMS driver only supports sending of Unicode strings. Reception of Unicode strings will still be interpreted as single byte strings.

The Command Agent does not recognise Multi-byte strings, so receiving any string in that format will not be accepted. It is therefore recommended to not use Hex mod when using the Command Agent in command reception mode.

The Command, Notify or String Agent can be used to send Unicode messages.



GSM_SMS : GSM2

Port Selection

☐ Enable Secondary

☒ Primary

☐ Secondary

Local Details

Port: COM6

Check used port(*) = used

Baud rate: 9600

Parity: None

Data bits: 8

Stop bits: 1

Retry: 3

Timeout: 5000

Modem Specific Adjustments

AT+CSMP="17,167,0,8"/r"

Device Utilization:

☐ Use with Command Agent

☐ Mark all scanned tags good (healthy) once on starting up

☐ Tags never grey out in a communication failure state

Suppress writes on startup for : 0 minutes.

Unicode Languages

☒ Use Unicode "HEX" mode

Unicode Code Page: 936 (ANSI/OEM - Simplified Chinese GBK)

Only 70 characters can be sent if using Unicode.

SMS Details

Modem type: Wavecom

Message centre number (optional):

PIN number (optional):

Delivery report separator: ,

SMS separator: ,

Incoming message polling interval (mSec): 0

Cost Control

☒ Enable burst protection cost control

Allow a maximum of 30 messages within a period of 60 minutes.

☐ Enable per month cost control

Maximum messages per month: 300

☐ Send warning SMS when cost control is in effect to :